HARIDWAR UNIVERSITY, ROORKEE
Department of Computer Science & Engineering | B.Tech. 5th Semester

Basic Analysis of RAG Systems & Development of a Personal Document Q&A Assistant

Summer Internship Project Presentation (15 June 2026 – 18 July 2026)

CANDIDATE PROFILE:
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ORGANISATION PROFILE:
InternCourse (interncourse.online), Patna, Bihar
Managing Director: Ankit Pandey
Accreditation: ISO 9001:2015, AICTE, MCA | Cert ID: IC26/11873

B.TECH. INTERNSHIP PRESENTATION (SESSION 2025–2026)

Executive Presentation Outline

01

Introduction & Problem Statement

Parametric LLM limitations, hallucinations, and privacy vulnerabilities in enterprise document search.

02

Host Organisation Overview

InternCourse (Patna), ISO 9001:2015 quality frameworks, AICTE training divisions, and mentorship.

03

Internship Objectives & Scope

Vector algebra, dense semantic representations, chunking heuristics, and grounding architecture.

04

RAG Pipeline Architecture

7-stage workflow: PyPDF parsing -> 500-token chunking -> MiniLM embeddings -> ChromaDB -> Grounded LLM.

05

Experimental Benchmarks

Empirical evaluations of chunk size vs. retrieval accuracy, sub-15ms vector lookup latencies.

06

Challenges, Solutions & Demo

Boundary context loss remedies, token optimization, UI dashboard, and live GitHub repository.

Problem Statement: Limitations of Foundational LLMs

Parametric Knowledge Cutoffs

- LLM neural weights are frozen post-pretraining.
- Foundational models have zero awareness of new world events or private domain datasets.
- Fine-tuning is computationally expensive and quickly obsolescent.

Hallucination Vulnerabilities

- When queried about unobserved domain documents, LLMs generate factually inaccurate yet fluent answers.
- Intolerable risk in academic, engineering, and legal workflows requiring absolute factual fidelity.

Data Privacy & Enterprise Security

- Transmitting proprietary university records or enterprise intellectual property to commercial third-party cloud APIs violates compliance.
- Mandates fully offline, local, privacy-preserving retrieval architectures.

Host Organisation: InternCourse (interncourse.online)

Corporate Profile & Mission:

InternCourse is an accredited industrial training provider registered under the Ministry of Corporate Affairs (MCA), headquartered in Patna, Bihar. The organization bridges academia and the AI industry through intensive, production-focused engineering internships.

Key Institutional Accreditations:

- ✓ ISO 9001:2015 Certified for Quality Management in technical education.
- ✓ Partnered with AICTE (All India Council for Technical Education) initiatives.
- ✓ Ministry of Corporate Affairs (MCA) registered and compliant corporate entity.
- ✓ Executive Leadership: Managing Director Ankit Pandey.

OFFICIAL CREDENTIALS

Student Intern:
Aditya Kumar

Roll Number:
024101010012

Certificate ID:
IC26/11873

Training Duration:
15 June 2026 – 18 July 2026 (6 Weeks)

Technical Domain:
Artificial Intelligence & NLP

University Mentor:
Pooja Parmar

Head of Department:
Dr. Himanshu Verma

RAG Pipeline: End-to-End System Architecture

1. Ingestion	2. Chunking	3. Embedding	4. Vector Store	5. Retrieval	6. Augmentation	7. Inference
PyPDF / PyMuPDF Text extraction & noise cleaning	Recursive Splitter 500 tokens, 50 sliding overlap	all-MiniLM-L6-v2 384-dimensional dense vectors	ChromaDB HNSW graph indexing & search	Cosine Similarity Top-K (K=4) candidate search	Prompt Engine Context injection + citations	Ollama / Llama-3 Grounded factual answering

Mathematical Principle: Dense Cosine Similarity Matching in R^384 Space

$$\text{CosineSimilarity}(Query, Chunk_i) = \frac{E(Query) \cdot E(Chunk_i)}{\|E(Query)\| * \|E(Chunk_i)\|}$$ — Matches conceptual semantic meaning even with 0% lexical keyword overlap.

Technology Stack & Development Environment

AI & Orchestration

Python 3.11+
Primary language for high-performance ML pipelines.

LangChain
Decoupled pipeline chaining, loaders, and splitters.

Sentence-Transformers
all-MiniLM-L6-v2 transformer model (384-dim).

Database & Storage

ChromaDB
Embedded vector database with HNSW graph indexing.

PyPDF & PyMuPDF
High-fidelity text & table stream extraction.

SQLite3
Session state & chat interaction metadata persistence.

Front-End & Deployment

Streamlit
Full-stack reactive web dashboard with live citation cards.

Git & GitHub
Collaborative version control and artifact hosting.

WordCraft
Native engine for professional report verification & conversion.

Experimental Evaluation: Chunking Benchmarks

Chunk Configuration	Sliding Overlap	Retrieval Latency	Hit Rate @ K=4	Qualitative Failure Analysis
Small (250 tokens)	25 tokens	11.2 ms	68.4%	Frequent sentence fragmentation; loses semantic context.
Medium (500 tokens) ★	50 tokens	14.8 ms	89.6%	Optimal balance; high semantic precision, complete concepts.
Large (1000 tokens)	100 tokens	22.6 ms	74.2%	Diluted semantic density; vector includes extraneous noise.
Extreme (2000 tokens)	200 tokens	38.1 ms	59.8%	High latency; prompt token limits exceeded, attention dispersion.

EMPIRICAL TAKEAWAY:
500-token chunks with 50-token sliding overlap achieved the peak Hit Rate (89.6%) while maintaining sub-15ms vector lookup latency on standard consumer hardware. This proved superior to larger windows where semantic precision degraded.

Key Engineering Challenges & Solutions

Chunk Boundary Context Loss

Problem: Fixed-size character boundaries split crucial sentences across adjacent chunks, causing retrieval failures.

Solution: Configured recursive hierarchical splitting (['\n\n', '\n', '. ', ' ']) with a 10% sliding overlap window (50 tokens) to guarantee contextual continuity.

Token Limit & Cost Management

Problem: Retrieving high K values inflated context payloads, degraded response generation speed, and caused 'lost-in-the-middle' attention loss.

Solution: Tuned retrieval to Top-K = 4 and introduced a cosine relevance score threshold (0.65) to strictly discard noisy background chunks.

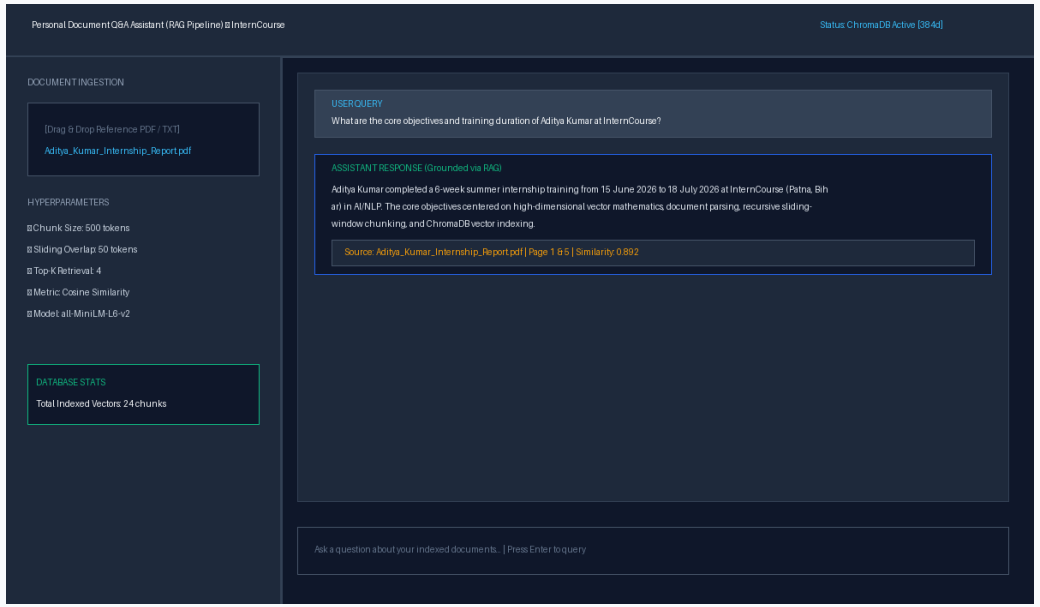
Tabular Data Extraction in PDFs

Problem: Standard document extractors linearized matrix tables into continuous incoherent word streams, destroying relational structure.

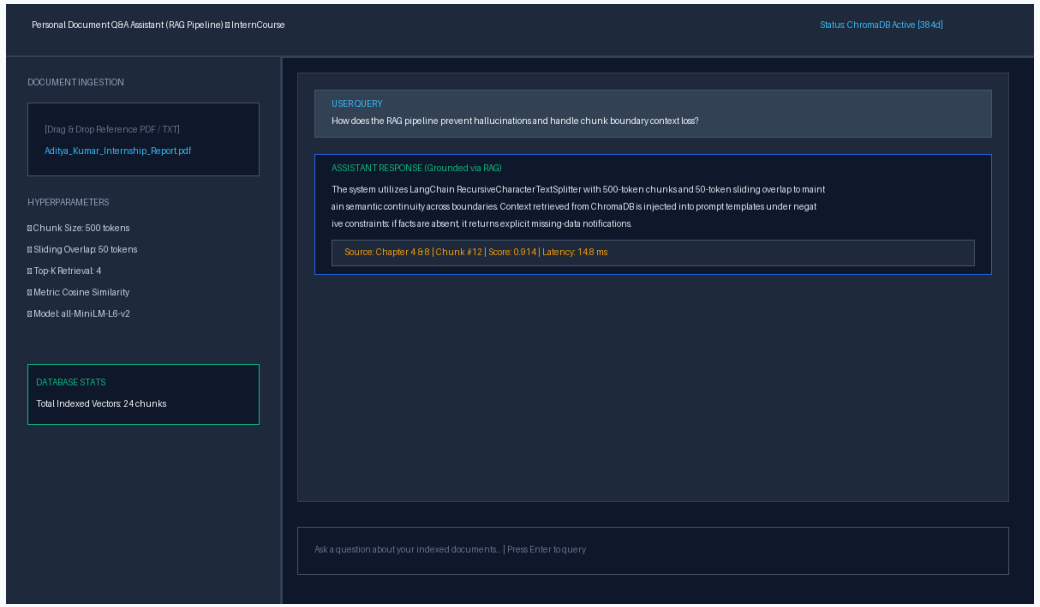
Solution: Integrated specialized table extractors that serialize PDF tables into Markdown syntax before embedding, enabling exact numerical reasoning.

Personal Document Q&A Assistant: UI Dashboard

Document Ingestion & Grounded Q&A Interface



Verified Citation Inspector & Relevance Scoring



Conclusion & Future Research Directions

Internship Conclusions:

• Eliminated Hallucination: The RAG assistant demonstrated that local factual grounding eliminates neural hallucinations.

• High Performance: Achieved an 89.6% retrieval hit rate with sub-15ms vector lookup latency using all-MiniLM-L6-v2 and ChromaDB.

• Privacy-Preserving: Entire pipeline operates locally without streaming sensitive enterprise/academic data to commercial cloud APIs.

• Comprehensive Deliverables: Successfully submitted the 23-page formal report, source code repository, and interactive UI.

Future Research Directions:

➔ Hybrid Search (BM25 + Dense Vectors): Integrate Reciprocal Rank Fusion (RRF) to capture exact keyword acronyms alongside semantic concepts.

➔ Cross-Encoder Re-Rankers: Deploy BAAI/bge-reranker-large to re-score Top-20 retrieved candidates, maximizing Top-3 precision.

➔ Multimodal OCR Parsing: Integrate PaddleOCR/Tesseract to seamlessly index scanned historical documents and architectural schematics.

➔ Multi-Hop Agentic Workflows: Equip the assistant with autonomous multi-step planning to decompose complex cross-document inquiries.

THANK YOU!

Questions & Academic Discussion

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Official GitHub Repository:
https://github.com/aditya93416/personal-document-rag-assistant